

STORMWISE DI² - SEALED ARTIFACT OUTPUT

FastPath 5.4 - FP-A01 (Artifact-Led Trust)

Tier 18 - Deterministic Work Product (DWP)

Status: FINAL - Scroll-Sealed / Canonical / Replayable

Artifact Type	Deterministic Weather Observation Work Product
Event Class	Spring Convection Boundary Test
Location	Chester County, PA Corridor
Date	2025-04-17
Final Tracking Value	13% (internal capacity index)
Operational Output	RAIN EVENT - NO

Executive Determination

A borderline spring convection environment was evaluated using multi-angle visual ingestion, application baselines, and radar validation. Elevated but insufficient convective structure was detected, followed by a deterministic downgrade after radar non-confirmation.

1. WORK PRODUCT SUMMARY

Summary:

A borderline spring convection environment was evaluated using multi-angle visual ingestion, application baselines, and radar validation. The system detected elevated but insufficient convective structure and enforced a deterministic downgrade after radar non-confirmation.

- Tracking Value: 13% (internal capacity index)
- Operational Output: RAIN EVENT - NO

2. INPUTS & ASSUMPTIONS

Inputs

- Multi-angle sky imagery (vehicle + ground)
- Application probability baselines (2% -> 5-7%)
- Radar state (no reflectivity)
- Human field observations: clustering behavior, vertical growth attempts, light variability, and repeated failed convective pulses

Assumptions

- Observation window bounded (~9:54-12:00 local)
- Rain defined as surface-contact precipitation at observation corridor
- Radar absence treated as hard constraint
- No external forcing introduced outside observed data

3. CONSTRAINTS APPLIED

- Protocol A (Deterministic Execution)
- Scrolls 0-166 (governance enforcement)
- Entropy Gatekeeper ($\Delta H \leq 0.0041$)
- DriftIndex Lock (0.000000)
- Cloud Morphology Constraint
- Spatial Clustering Constraint
- Vertical Growth Threshold Logic
- Radar Confirmation Requirement
- Cap Integrity Enforcement
- Binary Event Enforcement

Constraint Results

Constraint	Result
Morphology	PASS
Clustering	PASS
Vertical Growth	PARTIAL
Radar	FAIL
Cap Break	HOLD
Energy Continuity	FAIL

4. OUTPUT

Deterministic Analysis

Multi-angle ingestion confirms a non-committing convective field.

Observed Progression

- Initial: shallow cumulus, low instability
- Mid-phase: clustering + vertical probing
- Peak: congestus attempts without dominance
- Late-phase: decay and sky re-expansion

Detected Signals

- Localized lift pockets
- Intermittent vertical growth
- Directional imbalance
- Absence of sustained updraft

Constraint Resolution

- Insufficient vertical energy
- Shallow moisture engagement
- No organized forcing
- Radar non-confirmation

System Classification

Active Spring Convection - Non-Committing

5. VERIFICATION BLOCK

Artifact-ID	FP54-T18-DWP-SWDI2-2025-04-17-EXTON-001
Input-Hash	54c1e8b7a3d9f2c6e1a4b0d7c9e3f5a2b6d8c1e7f4a9b3c0d2e6f1a8b7c9d3
Output-Hash	e2a7c4d9f1b3a6c8d0e5f2b7c9a1d4e6f3b8a0c5d7e2f9a1b6c4d8e3f0a7b2
Rule-Layer	Tier 18 / FastPath 5.4
Entropy-Band	$\Delta H \leq 0.0041$
WP-Schema	DWP-5.4-v1
Replay-Class	Deterministic
Replay-Key	/replay/FP54-SWDI2-2025-04-17
Verification Endpoint	/verify/FP54-SWDI2-2025-04-17
Constraint-Stack-Hash	c9d3e7a1b5f2d8c4a6e0b3f7c1d9a2e6b4c8f0d5a7e3c1b9d6f2a4c8e1b7d3
Section Map	1-6 (DWP Compliant)

6. DETERMINISTIC EVIDENCE RECEIPT (DER)

Timestamp	2026-04-17T12:00:00Z
Input Classification	Multi-angle atmospheric observation + baseline comparison + field validation
Domain Routing	Weather -> Convection Threshold Assessment
Execution Stack	1. Morphology parsing 2. Clustering validation 3.

	Vertical growth detection 4. Multi-angle synthesis 5. Radar constraint enforcement 6. Threshold downgrade 7. Binary decision lock
Execution Path	Input -> Constraint Stack -> Deterministic Resolution -> DWP Seal
Entropy Confirmation	PASS ($\Delta H = 0.0000$)
DriftIndex	0.000000
Replay Eligibility	FULL

Final Conclusion

StormWise DI² under FastPath 5.4 Tier 18 execution produced a deterministic work product identifying a borderline convective environment with insufficient completion capacity. Despite observable instability signals, no sustained vertical dominance occurred, no cap break was achieved, radar provided no validation, and the precipitation pathway was invalidated. System behavior elevated the initial assessment based on structure, enforced constraint reconciliation, executed a deterministic downgrade, and finalized a binary output. Final Tracking Value: 13%. Final Operational Output: NO RAIN EVENT.

ARTIFACT STATUS: SEALED - DWP-COMPLIANT - CANONICAL - REPLAYABLE